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ELISA Protocol

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We use this protocol in our group and it is working.

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Abstract

ELISA (enzyme-linked immunosorbent assay) is a plate-based assay technique designed for detecting and quantifying peptides, proteins, antibodies, and hormones. In an ELISA, an antigen must be immobilized to a solid surface and then complexed with an antibody that is linked to an enzyme. Detection is accomplished by assessing the conjugated enzyme activity via incubation with a substrate to produce a measurable product. The most crucial element of the detection strategy is a highly specific antibody-antigen interaction.

This protocol describes the steps for performing sandwich ELISA.

Materials

STEP MATERIALS

 Mammal Cell Protein Extraction Reagent **Boster Bio Catalog #AR0103**

 Mammal Cell Protein Extraction Reagent **Boster Bio Catalog #AR0103**

Protocol materials

 Mammal Cell Protein Extraction Reagent **Boster Bio Catalog #AR0103**

Sample Preparation

- 1 The procedure below provides a general guide for preparing commonly tested samples used in ELISA assays. Please check the literature for experiments similar to yours for your new assay development. In this protocol, the following samples are covered:

1. Cell Culture Supernatant
2. Cell Culture (Conditioned) Media
3. Cell Lysate
4. Tissue Homogenate
5. Tissue (Bone)
6. Serum
7. Plasma
8. Urine
9. Saliva
10. Milk

Note

General tips for ELISA sample preparation:

- Serum, plasma, cell and tissue extracts are typically diluted by 50% with binding buffer.
- Total protein concentration of homogenate should be at least 1 mg/mL. However, 2 mg/mL or more would be better.
- The collected samples can be kept for different periods: 48 hours (2-8°C), 1 month (-20°C), or 6 months (-70°C).
- The protein concentration of your lysates can be determined by a total protein assay not inhibited by detergents such as the Bicinchoninic acid (BCA) assay. The volume of each sample can also be normalized to deliver the same amount of total protein for each assay.

Sample Preparation - Cell Culture Supernatant

- 2 Centrifuge cell culture media at 1,500 rpm and 4°C for 10 minutes.

 4 °C

- 3 Assay immediately or aliquot supernatant and hold at -80°C. Avoid freeze/thaw cycles.

 -80 °C

Sample Preparation - Cell Culture (Conditioned) Media

- 4 Since serum tends to contain cytokines which may produce significant background signals, we suggest the preparation of serum-free or low-serum medium samples. If it is



necessary to test serum containing medium, we also suggest running an uncultured medium blank to obtain the baseline signals which can then be subtracted from the cultured media sample data.

Sample Preparation - Cell Culture (Conditioned) Media - Day 1

- 5 Plate cells in complete growth media (with serum) until the desired level of confluence is achieved.

Note

The optimal number of seeded cells varies from one cell type to another and may need to be empirically determined. We suggest seeding around 1 million cells in 100 mm tissue culture plate with complete growth medium.

Sample Preparation - Cell Culture (Conditioned) Media - Day 4

- 6 Remove growth media and gently wash cells using 2-3 mL of warm PBS.
- 7 Repeat the wash step.
- 8 Remove PBS and replace the medium with serum-free or low serum containing medium (e.g. medium containing 0.2% calf serum).
- 9 Incubate for 1-2 days.

Sample Preparation - Cell Culture (Conditioned) Media - Day 6

- 10 Collect medium.
- 11 Centrifuge at 1,500 rpm and 4°C for 10 minutes.
 4 °C
- 12 Aliquot the supernatant and keep it at -80°C until experiment. Avoid freeze/thaw cycles. Most samples prepared this way can be stored for at least 1 year.

 -80 °C

Sample Preparation - Cell Lysate

- 13 Collect and rinse cells in PBS.
- 14 Homogenize and lyse cells thoroughly in lysis buffer (e.g. Mammal Cell Protein Extraction Reagent: **AR0103, Boster Bio**).

 Mammal Cell Protein Extraction Reagent **Boster Bio** Catalog #AR0103

Note

Guidelines on lysis buffer

- Avoid more than 0.1% SDS or other strongly denaturing detergents. In general, non-ionic detergents such as Triton X-100 or NP-40 are best. Zwitterionic detergents such as CHAPS or mild ionic detergents such as sodium deoxycholate will work.
- Do not use more than 2% v/v total detergent.
- Avoid the use of sodium azide.
- Avoid more than 10 mM reducing agents (e.g. dithiothreitol, mercaptoethanols).

- 15 Centrifuge cell lysate at approximately 10,000 x g and 4°C for 5 minutes.

 4 °C

- 16 Assay immediately or aliquot supernatant and hold at -80°C. Avoid freeze/thaw cycles.

 -80 °C

Sample Preparation - Tissue Homogenate

- 17 Rinse tissue with PBS to remove excess blood.
- 18 Chop tissue into 1-2 mm pieces on ice in ice-cold buffer. Keep on ice for immediate homogenization or at -80°C for later use.
- 19 Prepare the extraction buffer. It can be prepared ahead of time and stored at 4°C.
- 100 mM Tris, pH 7.4
 - 150 mM NaCl
 - 1 mM EGTA
 - 1 mM EDTA
 - 1% Triton X-100 0.5%
 - 0.5% sodium deoxycholate



- 20 Immediately before use, the extraction buffer must be supplemented with the following to generate a complete extraction buffer:
 - Phosphatase inhibitor cocktail (as directed by manufacturer)
 - Protease inhibitor cocktail (as directed by manufacturer)
 - PMSF (Phenyl Methyl Sulfonyl Floride) to 1 mM
- 21 For every 0.1 mg of tissue, add 500 μ L of complete extraction buffer to the tube and homogenize.
- 22 Rinse the blade of the homogenizer 2X with 500 μ L extraction buffer.
- 23 Place the sample on a shaker at 4°C for 1 hour.

 4 °C
- 24 Centrifuge the sample at approximately 10000 X g for 5 minutes.
- 25 Assay immediately or aliquot supernatant (soluble protein extract) and hold at -80°C. Avoid freeze/thaw cycles.

 -80 °C

Sample Preparation - Tissue (Bone)

- 26 Extract de-mineralized bone samples in 4M Guanidine-HCl and protease inhibitors.
- 27 Dissolve the final sample in 2M Guanidine-HCl.

Sample Preparation - Serum

- 28 Collect whole blood into a tube without additives.

Note

Hemolysis should be avoided while collecting serum samples. Samples that have undergone hemolysis may increase non-specific staining in HRP-conjugated ELISA assays.

- 29 Keep the blood at room temperature for 30 minutes.



30 Centrifuge at 3,000 rpm and 4°C for 10 minutes.

4 °C

31 Assay immediately or aliquot supernatant (serum) and hold at -80°C. Avoid freeze/thaw cycles.

-80 °C

Sample Preparation - Plasma

32 Collect whole blood into a tube containing anticoagulant. Different proteins may require different anticoagulants. See datasheet for details on which anticoagulant to use.

33 Centrifuge at 3,000 rpm and 4°C for 10 minutes.

4 °C

34 Assay immediately or aliquot supernatant (plasma) and hold at -80°C. Avoid freeze/thaw cycles.

-80 °C

Sample Preparation - Urine

35 Collect urine into a sterile or disposable container. Fresh urine samples must be used immediately or saved to avoid reproduction of bacteria which produce endogenous HRP that may potentially give false positive results.

36 Centrifuge sample at 10,000 x g for 1 minute.

37 Assay immediately or aliquot supernatant and hold at -80°C. Avoid freeze/thaw cycles.

-80 °C

Sample Preparation - Saliva

38 Collect saliva using a collection device without any protein binding or filtering capabilities (e.g. Salivette).

39 Centrifuge at 10,000 x g and 4°C for 2 minutes.

4 °C

40 Assay immediately or aliquot supernatant and hold at -80°C. Avoid freeze/thaw cycles.



-80 °C

Sample Preparation - Milk

41 Centrifuge at 1500 x g and 4°C for 15 minutes.

4 °C

42 Collect the aqueous fraction and repeat this process 3 times.

43 Filter through a 0.2 µm filter.

44 Assay immediately or aliquot supernatant and hold at -80°C. Avoid freeze/thaw cycles.

-80 °C

Reagent Preparation - Standard Solutions

- 45
- 10,000 pg/mL: Add 1 mL of sample diluent buffer into one tube of standard (10 ng per tube) and mix thoroughly. Store this solution at 4°C for up to 12 hours (or -20°C for 48 hours) and avoid freeze-thaw cycles.
 - 5,000 pg/mL: Mix 0.3 mL of 10,000 pg/mL with 0.3 mL of sample diluent buffer and mix thoroughly.
 - 2,500 pg/mL: Mix 0.3 mL of 5,000 pg/mL with 0.3 mL of sample diluent buffer and mix thoroughly.
 - Perform similar dilutions until the standard solutions with these concentrations (pg/mL) are made: 1250, 625, 312, 156, and 78.
 - Add 100 µL of each of the diluted standard solutions to the appropriate empty wells. Repeat in duplicate or triplicate for accuracy.

Note

The standard solutions are best used within 2 hours.

Reagent Preparation - Biotinylated Antibody

- 46
- Calculate the total volume needed for the assay by multiplying 0.1 mL/well and the number of wells required. Add 2-3 extra wells to the calculated number of wells to account for possible pipetting errors.

- Generate the required volume of diluted antibody by performing a 1:100 dilution (For each 1 μ L concentrated antibody, add 99 μ L antibody dilution buffer) and mixing thoroughly.

Reagent Preparation - Avidin-Biotin-Peroxidase Complex (ABC)

- 47
- Calculate the total volume needed for the assay by multiplying 0.1 mL/well and the number of wells required. Add 2-3 extra wells to the calculated number of wells to account for possible pipetting errors.
 - Generate the required volume of diluted ABC solution by performing a 1:100 dilution (For each 1 μ L concentrated ABC solution, add 99 μ L ABC dilution buffer) and mixing thoroughly.

Note

The diluted ABC solution should not be prepared more than 1 hour prior to the experiment.

Sandwich ELISA Protocol

- 48
- All of the **ELISA kits from Boster** use the sandwich format and biotin-streptavidin chemistry. Our ELISA assays require the dilutions of standard solutions, biotinylated antibody (detection antibody), and avidin-biotin-peroxidase complex.

Sandwich ELISA Protocol - Capture Antibody Coating (Not required if using Boster's pre-adsorbed PicoKine ELISA kits)

- 49
- Dilute the capture antibody to a final concentration of 1-10 μ g/mL in bicarbonate/carbonate antigen-coating buffer (100 mM NaHCO_3 in deionized water; pH adjusted to 9.6).
- 50
- Pipette 100 μ L of diluted antibody to each well of a microtiter plate.
- 51
- Cover the plate with adhesive plastic and incubate at 4°C overnight (or 37°C for 30 minutes).
- 52
- Remove the coating solution and wash the plate 3X with 200 μ L PBS (Phosphate Buffered Saline) buffer (10 mM Na_2HPO_4 and 1.8 mM NaH_2PO_4 in deionized water with 0.2% Tween 20; pH Adjusted to 7.4) for 5 minutes each time. The coating/washing solutions can be removed by flicking the plate over a sink. The remaining drops can be



removed by patting the plate on a paper towel or by aspiration. Do not allow the wells to dry out at any time.

Sandwich ELISA Protocol - Blocking (Not required if using Boster's pre-adsorbed PicoKine ELISA kits)

- 53 Pipette 200 μ L blocking buffer (5% w/v non-fat dry milk in PBS buffer) per well to block residual protein-binding sites. Alternatively, BSA or BlockACE can be used to replace non-fat dry milk.
- 54 Cover the plate with adhesive plastic and incubate for 1-2 hours at 37°C (or at 4°C overnight).
- 55 Remove the blocking solution and wash the plate 2X with 200 μ L PBS for 5 minutes each time. Flick the plate and pat the plate as described in the coating step.

Sandwich ELISA Protocol - Reagent Preparation

- 56 Prepare the diluted standard solutions, biotinylated antibody and ABC solutions as described in the above Reagent Preparation section.

⇒ [go to step #45](#) Standard Solutions Preparation

⇒ [go to step #46](#) Biotinylated Antibody Preparation

⇒ [go to step #47](#) ABC Solution Preparation

Sandwich ELISA Protocol - Sample (Antigen) Incubation

- 57 Serially dilute the sample with blocking buffer immediately before use. The optimal dilution should be determined by a titration assay according to the antibody manufacturer.
- 58 Pipette 100 μ L of each of the diluted sample solutions and control to each empty well. Repeat in duplicate or triplicate for accuracy. The negative control should be species- and isotype-matched as well as non-specific immunoglobulin diluted in PBS buffer.
- 59 Cover the plate with adhesive plastic and incubate for 2 hours at room temperature.
- 60 Remove the content in the wells and wash them 3X with 200 μ L PBS buffer for 5 minutes each time. Flick the plate and pat the plate as described in the coating step.

Sandwich ELISA Protocol - Biotinylated Antibody Incubation



- 61 Pipette 100 μL of diluted antibody to the wells with the control, the standard solutions, and the diluted samples.
- 62 Cover the plate with adhesive plastic and incubate for 1 hour at 37°C (or 2 hours at room temperature). These incubation times should be sufficient to receive a strong signal. However, if a weak signal is observed, perform incubation overnight at 4°C for a stronger signal.
- 63 Remove the content in the wells and wash them 3X with 200 μL PBS for 5 minutes each time. Flick the plate and pat the plate as described in the coating step.

Sandwich ELISA Protocol - ABC Incubation

- 64 Pipette 100 μL of diluted ABC solution to the wells with the control, the standard solutions, and the diluted samples.
- 65 Cover the plate with adhesive plastic and incubate for 0.5 hour at 37°C.
 37 °C
- 66 Remove the content in the wells and wash them 3X with 200 μL PBS buffer for 5 minutes each time. Flick the plate and pat the plate as described in the coating step.

Sandwich ELISA Protocol - Substrate Preparation

- 67 Prepare the substrate solution immediately before use or bring the pre-made substrate to room temperature. The two widely used enzymes for signal detection are horseradish peroxidase (HRP) and alkaline phosphatase (AP), and their corresponding substrates, stopping solutions, detection absorbance wavelengths and color developed are as follows:

	HRP	TMB	2M H_2SO_4	450	Yellow
	AP	pNPP	0.75M NaOH	405	Yellow

* TMB: 3,3',5,5'-tetramethylbenzidine; pNPP: p-nitrophenyl-phosphate

**Note**

- The TMB substrate must be kept at 37°C for 30 minutes before use.
- Hydrogen peroxide can also act as a substrate for HRP.
- Sodium azide is an inhibitor of HRP. Do not include the azide in buffers or wash solutions if HRP-labeled conjugate is used for detection.

Sandwich ELISA Protocol - Signal Detection

- 68 Pipette 90 μL of substrate solution to the wells with the control, the standard solutions, and the diluted samples.
- 69 Incubate the plate at 37°C in the dark. If TMB is used, shades of blue will be observed in the wells with the most concentrated solutions. Other wells may show no obvious color.
- 37 °C
- 70 Color should be developed in positive wells after 15 minutes. After sufficient color development, pipette 100 μL of stop solution to the appropriate wells (if necessary).
- 71 Read the absorbance (OD: Optical Density) of each well with a plate reader.

Sandwich ELISA Protocol - Data Analysis

- 72 Prepare a standard curve using the data produced from the diluted standard solutions. Use absorbance on the Y-axis (linear) and concentration on the X-axis (log scale).
- 73 Interpret the sample concentration from the standard curve.